

EVOLVING GEOAI FOR DISASTER RESILIENCE: FROM DISCRIMINATIVE DAMAGE
ASSESSMENT TO GENERATIVE MULTI-VIEW UNDERSTANDING AND AGENTIC DECISION
SUPPORT

A Dissertation Proposal

by
YIFAN YANG

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Chair of Committee, Dr. Lei Zou
Committee Members, Dr. Heng Cai
Dr. Andrew Klein
Dr. Zhengzhong Tu
Head of Department, Dr. Oliver Frauenfeld

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Project Summary

Overview

Disaster resilience is a key geographical issue because it is rooted in the spatial distribution of vulnerability, exposure variations, and spatiotemporal evolutionary processes shaped by human-environment systems. Specifically, when a disaster occurs, our focus as geographers lies not only in what happened, but more importantly in where it occurred, the severity of damage across different locations, who was affected, and how to allocate limited response and recovery resources spatially. Overhead imagery from satellites, aircraft, and drones offers broad coverage but is limited to viewing the built environment from above; conversely, ground-level imagery better reflects the ruins, structural damage, and blocked roads people actually experience, yet it suffers from spatial incompleteness and temporal inconsistency. Computer vision has enabled us to classify damage from any viewpoint; however, these models are often trained on isolated events, leading to misleading priority rankings based on judgments that appear confident yet are actually incorrect. Furthermore, they stop at category labels and fail to provide the site-specific explanations and decision-making rationale that emergency responders, residents, and planners truly need.

This study traces and advances the evolution of Geospatial Artificial Intelligence (GeoAI) for disaster resilience, moving through three stages of progressively enhanced capabilities: discriminative models, generative models, and agentic models. Centered on seven interconnected case studies spanning these three stages, the research: (1) begins with discriminative damage assessment, enhancing the completeness and reliability of observations for a single hurricane, despite limited ground-level perspectives, by pairing pre- and post-disaster street-view imagery and arbitrating between visual models and vision-language models; (2) advances to generative multi-view understanding, employing a reproducible, zero-shot multi-agent workflow to jointly reason across regions and disaster types using satellite and street-view evidence, synthesizing missing ground-level views from satellite imagery via generative models, and organizing the roles of generative AI across disaster phases within a full-lifecycle framework; and (3) ultimately moves toward agentic decision support, predicting street-level damage using pre-event information and developing RAY (Resilience Assistant for You), an accountable GeoAI agent that ensures the chain of evidence. Each stage builds upon the data, models, and lessons learned from the preceding one.

Intellectual Merit

This research advances disaster GeoAI across three stages: first, the discrimination stage re-evaluates multi-perspective, multi-temporal, and multi-modal observations by treating model discrepancies and baseline data as vital evidence, prioritizing reliability over mere accuracy; second, the generation stage employs a reproducible zero-shot multi-agent architecture, divided into specialized perception, restoration, and reasoning roles, to achieve structured cross-modal analysis without event-specific fine-tuning; and finally, the agentic decision support stage establishes accountability as a core design principle, requiring autonomous agents to link spatial judgments to verifiable hashed evidence, explicitly declare unknowns, and abstain when unsupported.

Broader Impacts

This research supports faster and more equitable emergency response, more trustworthy use of generative AI in high-stakes settings, and broader public access to place-based disaster intelligence. Hyperlocal, reliability-aware damage maps can help agencies direct search, rescue, and debris operations to the hardest-hit streets in the first hours after an event and can shorten the property-level inspections that delay insurance and recovery. Zero-shot, cross-view pipelines lower the barrier for communities that lack event-specific training data or specialist analysts. Pre-event damage prediction turns the same models toward preparedness, and the RAY system delivers dossiers and planning trade-offs to residents and decision-makers, with attached evidence lineage.

Project Description

1. Introduction and Significance

Disasters often exhibit uneven spatial distribution; for instance, two neighborhoods separated by only a single block may experience vastly different levels of damage. The value of disaster assessment lies in capturing these disparities and translating them into actionable information—such as rapid post-disaster mapping and damage reports. In the immediate aftermath of a disaster, emergency responders must determine where to prioritize their efforts and on what evidence to base those decisions; in the ensuing weeks, agencies must allocate limited personnel, supplies, and funds; and in the years that follow, site-specific assessment data underpin insurance claims, reconstruction, and resilience planning (Cutter et al., 2008; Ghaffarian et al., 2018). Given the spatial nature, time sensitivity, and significant impact of assessment activities across these stages, disaster damage assessment is fundamentally a geographical issue, whereas processing disaster imagery is a computer vision problem.

Regarding disaster imagery, remote sensing technology has long provided an aerial perspective on disaster impacts. Satellite, aerial, drone, and LiDAR imagery have been widely used to map flood extents, roof damage, debris distribution, and burned structures following hurricanes, earthquakes, and wildfires (Hauptman et al., 2024; Kerle et al., 2020; Vetrivel et al., 2018); furthermore, the emergence of benchmark datasets such as xBD has established building damage classification based on nadir imagery as a standard deep learning task (Gupta et al., 2019). While this perspective offers broad coverage, it is limited to roofs and building footprints. Ground-level imagery (such as street-view images), by contrast, aligns more closely with human perceptions of disasters, capturing details like wall damage, fallen trees, and obstructed access points (Anguelov et al., 2010; Biljecki & Ito, 2021; Zhai & Peng, 2020). However, ground-level imagery has incomplete spatial coverage and often lacks temporal continuity, as obtaining pre- and post-disaster imagery for the same location is often difficult. Each perspective captures only one facet of the disaster's impact.

GeoAI, defined as spatially explicit artificial intelligence that leverages geographic data and spatial context to generate geographic information, knowledge, and services, provides the methodological engine for the intelligence required to interpret such imagery (Janowicz et al., 2020; Li & Hsu, 2022). Notably, GeoAI itself is continuously evolving. Discriminative models address closed-ended questions with labeled answers (such as whether a building is damaged and the extent of that damage), achieving extremely high accuracy when labeled data is available (Cheng et al., 2021; Nex et al., 2019). In contrast, generative models and vision-language models can answer open-ended and constructive questions: why a location appears severely damaged, what evidence supports that assessment, what elements are missing from the scene, and what actions emergency responders should take next (Achiam et al., 2023; Mai et al., 2024; Radford et al., 2021). Recently, agentic systems built upon these models have gained the ability to plan analytical tasks, invoke spatial analysis tools, and interact with users (Li & Ning, 2023). This paper views these developments as successive stages in the evolution of GeoAI for disaster resilience, a process through which disaster intelligence becomes more comprehensive, more transferable, and ultimately, more accountable.

Viewed through this lens, three challenges remain. First, place is only partially observed. Most damage models rely on a single post-event view of a single event, discard the pre-disaster baseline, and treat disagreement between views or between models as noise rather than as information. Second, place-specific disasters require rapid assessment without event-specific training. Each event unfolds in a unique local context, yet models trained on one hurricane transfer poorly to another, and responders cannot wait for new labels, retraining, and validation. Accuracy alone is also insufficient in this setting, because modern networks are systematically overconfident and a confidently wrong severity label can misdirect a rescue team (Guo et al., 2017). Third, place is difficult to anticipate, inspect, and communicate. Disaster intelligence should help people understand possible impacts before events and inspect affected environments afterward; however, current pipelines end with a

label or a map, and the emerging generation of autonomous GeoAI agents that could deliver richer intelligence lacks mechanisms for accountability, evidence lineage, and refusal when evidence is insufficient.

This dissertation addresses these challenges through seven linked case studies organized into three components that follow GeoAI's evolution. The first component, the discrimination stage, uses pre-trained vision models on bi-temporal street-view imagery and multimodal arbitration to observe a single hurricane more completely and more reliably. The second component, the generation stage, uses generative AI: a reproducible multi-agent workflow and a lifecycle framework assess damage rapidly and interpretably across views, regions, and hazard types without event-specific training, and generative models synthesize missing ground-level views. The third component, the agentic stage, turns the resulting intelligence toward preparedness and decision support through pre-event damage prediction and an accountable GeoAI agent. Each stage inherits the data, models, and lessons of the previous one, so that together these studies show how progressively more capable AI lets us see, describe, and govern disasters at the scale of streets and neighborhoods.

2. Literature Review and Conceptual Framework

2.1 Discriminative damage assessment: observing place across views, scales, and times

Deep learning has transformed remote sensing for disaster response (Ma et al., 2019; Zhu et al., 2017). Convolutional networks and, more recently, vision transformers classify building damage from very-high-resolution satellite imagery (Gupta et al., 2019; Xu et al., 2019), segment flooded structures by fusing multi-resolution and multi-temporal sources (Rudner et al., 2019), and detect structural failures in oblique aerial and drone imagery (Kerle et al., 2020; Vetrivel et al., 2018). Post-hurricane aerial assessment has reached operational accuracy for coarse categories (Cheng et al., 2021). Nevertheless, overhead views cannot see façades, ground-floor flooding, or debris under tree canopy, and their damage classes are correspondingly coarse.

Street-view imagery fills part of this gap. Reviews document its rapid adoption in urban analytics and public health (Biljecki & Ito, 2021; Kang et al., 2020), and early disaster studies used Google Street View to document hurricane damage at the parcel level (Zhai & Peng, 2020). Multimodal models that combine street-view images with structured building data improve post-hurricane assessment (Zhang et al., 2026), and cross-view geolocalization now links ground photographs to satellite tiles for disaster mapping (Li et al., 2025). Yet almost all street-view damage models use only post-event images, so they mistake pre-existing conditions for impacts and temporary debris for structural failure. Moreover, accuracy-centered evaluation hides a second problem: modern neural networks are poorly calibrated, and their most confident predictions are not necessarily their most correct ones (Guo et al., 2017). In disaster response, an overconfident error is operationally worse than an uncertain one, because a low-confidence prediction can be flagged for human review while a confident one is acted upon. Contrastive vision-language models such as Contrastive Language-Image Pre-training (CLIP) reason about scenes through natural-language descriptions and exhibit a different error profile than purely visual classifiers (Radford et al., 2021), suggesting that combining, rather than replacing, model families can improve reliability. Interpretability tools such as Gradient-weighted Class Activation Mapping (Grad-CAM) can confirm whether a classifier attends to real damage evidence rather than background (Selvaraju et al., 2017), and interpretability is increasingly regarded as a prerequisite for deploying machine learning in consequential settings (Doshi-Velez & Kim, 2017). The discrimination stage of this dissertation therefore asks how baselines, model disagreement, and interpretability can make a single, incomplete ground-level view trustworthy.

2.2 Generative multi-view understanding: assessing across views, regions, and hazards

Large language models and multimodal foundation models open a further path. Foundation models transfer across tasks with few or no labels (Bommasani et al., 2021; Mai et al., 2024), and multimodal models can describe a damaged scene, weigh evidence from different views, and generate reports for human users (Achiam et al., 2023). Applied to disasters, this promises zero-shot assessment that does not require event-specific training and can extend from one hurricane to many events and hazard

types. Generative vision models add a complementary capability: when post-event street-level coverage is missing altogether, cross-view image synthesis from satellite to ground, which has matured in computer vision through conditional generative adversarial networks, geometry-guided panorama synthesis, and diffusion models (Ho et al., 2020; Regmi & Borji, 2018; Rombach et al., 2022; Shi et al., 2022), could reconstruct a plausible ground-level view of the same place.

Three limitations remain. First, results are typically reported for one model on one dataset, so reproducibility across models, regions, and hazards is unknown. Second, generated reasoning is rarely checked for consistency with the underlying imagery, so fluent but unsupported explanations pass unnoticed; likewise, cross-view synthesis has not been adapted to represent disaster damage or evaluated for trustworthiness beyond pixel-level realism. Third, generative AI is usually studied for a single phase of the disaster cycle, leaving open how it should be organized across preparedness, response, recovery, and mitigation. This dissertation's generation stage responds to these limitations with a reproducible multi-agent workflow, a structure-aware evaluation of synthesized views, and a lifecycle framework.

2.3 Agentic decision support: from assessment to preparedness and accountable delivery

Place-based resilience frameworks emphasize that assessment should inform decisions before, during, and after events (Cutter et al., 2008). Two developments make this feasible for damage assessment. Earth-observation foundation models such as AlphaEarth now provide globally consistent 10-m embeddings that transfer to diverse mapping tasks with sparse labels (Brown et al., 2025), offering a scalable pre-event description of the physical environment that can be combined with hazard forecasts and street-level imagery. At the same time, autonomous GIS and GeoAI agents built on large language models can plan analyses, call spatial tools, and converse with users (Li & Ning, 2023), suggesting a delivery layer in which residents and planners query disaster intelligence directly. Autonomy, however, raises the accountability problems long recognized in AI safety: agents can assert claims without evidence, act outside their competence, and leave no trace of how they reached a conclusion (Amodei et al., 2016; Doshi-Velez & Kim, 2017). What is missing is a GeoAI architecture in which every spatial claim carries its evidence lineage, unknowns are declared rather than filled in, and the system refuses to answer when evidence is insufficient. The agentic stage of this dissertation supplies that architecture.

2.4 Conceptual Framework

Building on these gaps, this dissertation develops a conceptual framework in which GeoAI for disaster resilience evolves through three stages, each responding to one of the three challenges and each building on the last (Figure 1). The discrimination stage responds to the partial observation of place: pre-trained vision models are applied to a single hurricane from a single ground-level perspective, and observation becomes more complete and reliable by pairing pre- and post-disaster street-view imagery and quantifying and arbitrating model uncertainty. The generation stage responds to the need for rapid, place-specific assessment without new labels: structured zero-shot reasoning by multimodal agents replaces event-specific training, extends assessment from one view and one event to satellite and street-view evidence across regions and hazard types, produces interpretable reports, synthesizes ground-level views where they are missing, and is situated within the full disaster lifecycle. The agentic stage responds to the difficulty of anticipating, inspecting, and communicating place-based impacts: damage is predicted before events, and intelligence is delivered through an accountable GeoAI agent that ties every claim to its evidence and abstains when evidence is insufficient.

The research plan consists of seven case studies organized into three components. The first component includes two discriminative studies: bi-temporal street-view damage assessment with pre-trained vision models, and a multimodal arbitration framework that improves reliability. The second component comprises three generative studies: RAPID, a reproducible multi-agent workflow for interpretable damage assessment from satellite and street-view imagery across hurricanes and wildfires; generative satellite-to-street synthesis of post-disaster views; and a lifecycle framework for

generative AI in disaster resilience. The third component develops two agentic studies: a preparedness-oriented model that predicts and visualizes street-level impacts from pre-event information, and RAY (Resilience Assistant for You), an accountable GeoAI agent for place-based disaster intelligence. Together, these studies move from classifying damage in one place and one view, to understanding damage across views, regions, and hazards, to deciding and communicating with accountable agents.

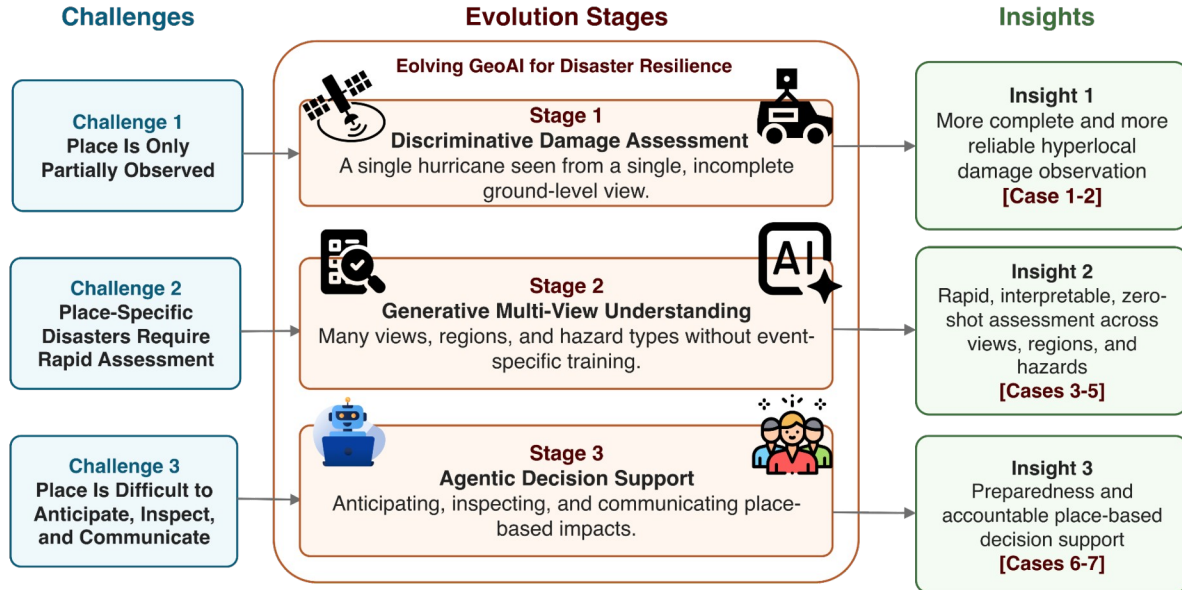


Figure 1. Conceptual framework connecting three challenges in disaster damage assessment to the discriminative, generative, and agentic stages of evolving GeoAI for disaster resilience (Case studies 1–7).

3. Research Questions and Hypotheses

Central Question. How does the evolution of GeoAI from discriminative to generative to agentic models make disaster damage assessment more complete, more rapid, and more reliable across views, regions, and hazards, and more useful for preparedness and decision support at the hyperlocal scale?

RQ1: Discriminative Damage Assessment (Case studies 1–2). Can pre-disaster baselines and arbitration between visual models and vision-language models make hyperlocal damage observation of a single hurricane more complete and more reliable?

H1. Pairing pre- and post-disaster street-view imagery will improve damage classification over post-event-only models and shift model attention toward true changes, and combining complementary model families will raise accuracy while reducing the share of overconfident errors.

RQ2: Generative Multi-View Understanding (Case studies 3–5). Can generative AI extend damage assessment from one view and one event to satellite and street-view evidence across regions and hazard types, through zero-shot multi-agent reasoning, synthesis of missing ground-level views, and a lifecycle organization of generative capabilities?

H2. Without event-specific training, structured multi-agent reasoning will achieve severity accuracy comparable to supervised baselines across hurricane and wildfire events, will produce reports whose stated evidence is consistent with the imagery, and will remain reproducible across model families; satellite-conditioned generative models will produce post-disaster street views whose structure and damage semantics are consistent with the overhead evidence; and the actionability of generated intelligence will vary systematically by disaster phase.

RQ3: Agentic Decision Support (Case studies 6–7). To what extent can street-level damage be anticipated from information available before an event, and can an accountable GeoAI agent deliver

place-based disaster intelligence to residents and planners with evidence lineage and explicit refusal boundaries?

H3. Pre-disaster street-view imagery, Earth-observation embeddings, hazard forecasts, and building context will predict post-event street-level severity well above chance under spatially blocked validation, and an agent architecture that ties every claim to hashed evidence and declared unknowns will outperform an unconstrained agent on claim support, refusal correctness, and user trust without sacrificing usefulness.

4. Research Design and Methods

4.1 Overall Research Design

This dissertation comprises seven studies organized into three components that correspond to three generations of GeoAI. Studies 1 and 2 are discriminative: Study 1 pairs pre- and post-disaster street-view imagery of a single hurricane, and Study 2 arbitrates between an image-only and an image-text model to improve reliability. Studies 3, 4, and 5 are generative: Study 3 builds and validates RAPID, a zero-shot multi-agent workflow over satellite and street-view imagery across hurricane and wildfire events; Study 4 synthesizes post-disaster street views from satellite imagery where ground-level coverage is missing; and Study 5 situates such generative capabilities within a lifecycle framework and evaluates their actionability. Studies 6 and 7 are agentic: Study 6 predicts street-level damage from pre-event information, and Study 7 develops and evaluates RAY, an accountable GeoAI agent. Table 1 summarizes the mapping between components, case studies, and research questions.

Table 1. Research components, case studies, and research questions

Component	Case studies	Research question addressed
Discriminative damage assessment	Case studies 1–2: Bi-temporal street-view assessment and multimodal arbitration for reliability	RQ1: Can pre-disaster baselines and multimodal arbitration make hyperlocal damage observation of a single hurricane more complete and more reliable?
Generative multi-view understanding	Case studies 3–5: RAPID multi-agent pipeline, satellite-to-street synthesis, and generative AI across the disaster lifecycle	RQ2: Can generative AI extend assessment across views, regions, and hazards through zero-shot reasoning, view synthesis, and lifecycle organization?
Agentic decision support	Case studies 6–7: Pre-event street-level damage prediction and the RAY accountable GeoAI agent	RQ3: Can pre-event information anticipate street-level damage, and can an accountable GeoAI agent deliver place-based intelligence with evidence lineage?

4.2 Component 1: Discriminative damage assessment (Case studies 1–2)

4.2.1 Study Context and Data

Case Study 1 examines Horseshoe Beach, Florida, a coastal community severely affected by Hurricane Milton in October 2024. We collected post-disaster street-level imagery at the local road level with a vehicle-mounted Mosaic 51 camera system, and obtained pre-disaster imagery for the same road segments from Google Street View. Images were geographically paired and labeled by trained annotators into mild, moderate, or severe damage, yielding 2,556 pre/post pairs; the resulting Milton bi-temporal dataset and study-area boundary have been released publicly. This is, deliberately, the narrowest setting in the dissertation: one hurricane, one community, and one ground-level

perspective that is itself incomplete. Case Study 2 uses the post-disaster images of the same dataset together with natural-language scene descriptions generated by a large language model and, for comparison, written by human annotators.

4.2.2 Measurement and Analysis

Case Study 1 compares three model settings: post-disaster images only, pre-disaster images only as a control, and paired pre- and post-disaster images. Pre-trained Swin Transformer and ConvNeXt backbones (Liu et al., 2021; Liu et al., 2022) are fine-tuned under several bi-temporal designs, including six-channel stacking, cross-attention, Siamese feature difference, and dual-channel feature fusion, and are evaluated with accuracy and class-wise F1. Grad-CAM heatmaps (Selvaraju et al., 2017) test whether models attend to true pre/post changes rather than background. Completed results show that accuracy improved from 66.14% for the single-image Swin baseline to 77.11% for the dual-channel feature-fusion ConvNeXt, and that pre-disaster context concentrates attention on genuine changes. In other words, the pre-disaster baseline is not merely additional data; it shifts the task from recognizing damage to recognizing change.

Case Study 2 shifts the evaluation from accuracy to reliability. An image-only vision transformer (Dosovitskiy et al., 2021), a text-only classifier, and a CLIP-based image-text model (Radford et al., 2021) are compared not only on accuracy but on a confidence-stratified error profile that separates overconfident, medium-confidence, and ambiguous errors. DamageArbiter accepts a prediction when the visual and image-text models agree and, when they disagree, arbitrates between them with a lightweight logistic-regression meta-classifier that uses only label-free confidence features, so that it can be deployed on unlabeled imagery. We assess performance with accuracy, weighted F1, and Matthews correlation coefficient under out-of-fold cross-validation, and map results to compare predicted and ground-truth severity spatially, because reliability must be judged where errors occur, not only in aggregate. Completed results show that the vision-only model produced overconfident errors in about 71% of its mistakes whereas the image-text model produced none, and that arbitration raised accuracy to 75.9% with MCC of 0.62 while sharply reducing overconfident errors.

4.3 Component 2: Generative multi-view understanding (Case studies 3–5)

4.3.1 Study Context and Data

The generation stage widens the setting along every dimension that the discrimination stage held fixed. Case Study 3 assembles three dataset types that span views, hazards, and regions: a cross-view set pairing satellite and street-view imagery for Hurricane Ian in Florida, the Milton bi-temporal street-view set from Case Study 1, and a wildfire set for the January 2025 Palisades and Eaton fires in Los Angeles County, in which street-level images are matched to structure-level damage inspections published by CAL FIRE. Case Study 4 draws on co-located satellite and street-view imagery from Hurricane Ian (2022) and Hurricane Milton (2024), aligned through geographic coordinates, to learn the mapping from an overhead view to a ground-level view of the same damaged place. Case Study 5 reuses these data to test generative models across the disaster lifecycle and additionally draws on publicly available emergency-management documents to define preparedness and recovery tasks.

4.3.2 Measurement and Analysis

RAPID is organized in three stages. Stage 1 prepares and aligns imagery around common locations. Stage 2 runs an autonomous multi-agent framework divided into specialized roles: a Disaster Perception Agent identifies hazard type and initializes the task; an Image Restoration Agent detects and repairs low-quality imagery using a quality score that combines contrast, sharpness, and a no-reference quality index and accepts a restoration only if quality improves; a Damage Recognition Agent identifies damaged objects and classifies severity from remote sensing and street-view inputs; and a Disaster Reasoning Agent analyzes impacts and generates a structured report. Stage 3 validates the outputs. All prompts are versioned and hashed so that runs can be reproduced, and the workflow is executed with multiple commercial and open multimodal models to test model dependence. We evaluate severity predictions with accuracy and a normalized cross-severity error that penalizes distant misclassifications more heavily; we evaluate generated reports by checking

each stated claim against the imagery as consistent, contradictory, or unsupported, and by a four-dimension rubric applied by both a language-model judge and human evaluators. Completed results show zero-shot severity accuracy between roughly 0.4 and 0.6 across models and events, with the strongest models approaching supervised baselines and restoration improving image quality on both views. This work has been accepted at ACM SIGSPATIAL 2026.

Case Study 4 trains conditional generative vision models based on latent diffusion (Rombach et al., 2022) to synthesize post-disaster street views from satellite imagery. Two mechanisms inject disaster semantics: vision-language-model-guided prompts describing the overhead damage, and a mixture-of-experts decoder specialized by severity level. Because pixel metrics such as FID reward realism rather than correctness, a structure-aware evaluation framework compares the synthesized views with reference imagery on geometric consistency, semantic segmentation agreement, and severity classification by an independent damage model, with human raters judging plausibility. This work has been accepted for presentation at the 2026 IEEE International Geoscience and Remote Sensing Symposium.

Case Study 5 develops a lifecycle framework that maps generative-AI capabilities, including description, restoration, synthesis, reasoning, and report generation, onto preparedness, response, recovery, and mitigation tasks, and evaluates a case study spanning cross-view and bi-temporal assessment. Beyond accuracy, outputs are scored for actionability by domain evaluators on a five-point rubric, and the analysis identifies where generated intelligence is informative but not yet actionable. This gap between informative and actionable is precisely what motivates the agentic stage. The chapter is under review for an edited volume on geography in the age of generative AI.

4.4 Component 3: Agentic decision support (Case studies 6–7)

4.4.1 Study Context and Data

Case Study 6 uses the Milton bi-temporal dataset as ground truth for post-event street-level severity and assembles the information that would have been available before landfall: pre-disaster street-view imagery, 64-dimensional AlphaEarth embeddings at 10-m resolution (Brown et al., 2025), National Hurricane Center wind and storm-surge forecasts issued before landfall, building footprints and parcel attributes, elevation, and American Community Survey estimates. Hurricane Ian and the Eaton Fire provide transfer tests. Case Study 7 builds RAY on three tiers of data: a continuously updated watch of active United States hazards from USGS, National Weather Service, National Hurricane Center, and interagency wildfire feeds; deep case dossiers for Hurricane Milton and the Eaton Fire that integrate the damage products of Case Studies 1 to 6 with population, critical-facility, and social-vulnerability layers; and cross-view evidence from the assessment workflows. Every input is snapshotted with a cryptographic hash so that its provenance can be verified.

4.4.2 Measurement and Analysis

Case Study 6 frames preparedness as prediction: a multimodal model encodes the pre-disaster street view, the AlphaEarth embedding of the surrounding cell, and tabular hazard and building features, and predicts the post-event severity class and a continuous risk score. Validation uses spatially blocked folds and a cross-event transfer test so that performance reflects generalization to unseen neighborhoods rather than spatial autocorrelation. The risk scores are aggregated into street-segment priority maps, and the generative model from Case Study 4 is conditioned on the predicted severity to visualize plausible pre-event post-disaster scenes, which are evaluated with the same structure-aware framework.

Case Study 7 implements RAY as a static-first progressive web application with an artifact-driven pipeline and a thin gateway around a large multimodal model. The system has a data plane, an agent plane, and a presentation plane, and is governed by a Steward Harness that enforces a policy of default deny: the agent may assert a claim about a place only if the claim cites registered evidence, unknowns are declared rather than inferred, and answers that would require unavailable evidence are refused with an explanation. A resident mode returns an address-level resilience dossier with citation click-through, and a planner mode exposes trade-off sliders, a lineage viewer, and validity badges that report which checks each product passed. Evaluation combines a verifiable test environment of

adversarial prompts and a policy matrix, measures of claim support and refusal correctness, and a user study with responders, planners, and residents comparing RAY with an unconstrained agent on trust, task completion, and comprehension.

5. Intellectual Merit

This dissertation makes three theoretical contributions to GeoAI for disaster resilience, one at each stage of the discriminative-to-generative-to-agentic trajectory.

(1) **In the discrimination stage, it reconceptualizes damage observation as a multi-perspective, multi-temporal, and multi-modal problem in which partial observation is the norm.** No single perspective or moment sees a damaged place completely, so pre-disaster baselines and disagreement between complementary models are treated as vital evidence about the place rather than as noise to be discarded. This perspective explains why change, reliability, and completeness must be measured together, and why reliability should take priority over mere accuracy.

(2) **In the generation stage, it redefines what counts as a good assessment model when disasters are time-sensitive and place-specific.** By dividing assessment into specialized perception, restoration, recognition, and reasoning roles; replacing event-specific fine-tuning with reproducible zero-shot reasoning that produces inspectable reports; and synthesizing missing views, the research shifts evaluation from accuracy on one event to reliability, interpretability, and transferability across views, regions, and hazards.

(3) **In the agentic stage, it establishes accountability as a core design principle of GeoAI systems.** Instead of treating autonomy and trust as opposites, the research formalizes how spatial judgments should be linked to verifiable hashed evidence, declared unknowns, and explicit refusal boundaries, and shows that place-based intelligence can be delivered to non-experts while remaining auditable. Human decision makers therefore remain the governors of disaster intelligence rather than its passive recipients.

6. Broader Impacts

The broader impacts of this dissertation lie in faster and fairer response, trustworthy use of generative AI in high-stakes settings, and wider public access to disaster intelligence.

(1) **It supports faster and more equitable emergency response and recovery.** Hyperlocal, reliability-aware damage maps can direct search, rescue, and debris operations to the hardest-hit streets within hours and can shorten the property-level inspections that delay insurance payouts and rebuilding, with the largest benefits for communities that lack the resources for rapid manual assessment.

(2) **It advances the responsible use of generative AI for disasters.** Zero-shot cross-view workflows lower the barrier for agencies without event-specific training data, while the reliability diagnostics, consistency checks, and accountability mechanisms developed here provide safeguards that transfer to other consequential applications of foundation models in public health, climate adaptation, and urban planning.

(3) **It broadens public access to place-based disaster intelligence.** Pre-event damage prediction gives residents and planners a concrete picture of possible impacts before a storm or fire, and RAY delivers dossiers and planning trade-offs with evidence attached. The datasets, prompts, code, and design documents from the completed studies are released openly, and the accountable-agent template is documented so that other GeoAI systems can adopt it.

7. Current Status and Timeline

The dissertation is well advanced. Case studies 1 through 5 have been completed. Case study 1 was published in *Computers, Environment and Urban Systems*; Case study 2 received a Best Student Paper award at the 2025 International Cartographic Conference and its extended version is under review at *Natural Hazards*; Case study 3 has been accepted at ACM SIGSPATIAL 2026, with a

companion abstract accepted for the CaGIS 2026 student paper competition; Case study 4 has been accepted at IEEE IGARSS 2026; and Case study 5 has been submitted as a chapter to an edited volume on geography in the age of generative AI. Together, these studies complete the discrimination and generation stages and establish the bi-temporal, arbitration, multi-agent, and synthesis workflows on which the agentic stage builds, and the Milton bi-temporal dataset, prompts, and workflow code have been released publicly.

Case study 7 is approximately 40% complete. The RAY prototype, including the live hazard watch, the Milton and Eaton deep-case dossiers, the Steward Harness with its policy, claim, and distribution rules, and the resident and planner interfaces, has been implemented, deployed, and covered by an automated evaluation environment of more than 350 tests; a short paper and demonstration have been submitted. The remaining work is the user study and the integration of the cross-view evidence tier. Case study 6 is about 20% complete: the pre-event data have been assembled, and AlphaEarth embeddings have been extracted for the study areas; model training, spatially blocked validation, and scenario visualization remain.

As shown in Figure 2, the preliminary examination was completed in June 2026, and the proposal defense is planned for Fall 2026. Case studies 6 and 7 will be completed between Fall 2026 and Spring 2027, with manuscripts submitted as each concludes. Dissertation integration and writing have already begun and will proceed alongside the remaining analyses, so that the complete dissertation can be circulated to the committee, revised, defended, and submitted in Spring 2027.

Tasks	Fall 2024	Spring 2025	Fall 2025	Spring 2026	Fall 2026	Spring 2027
Case Study 1	Done					
Case Study 2		Done				
Case Study 3			Done			
Case Study 4				Done		
Case Study 5				Done		
Case Study 6					In Progress (20%)	
Case Study 7					In Progress (40%)	
Prelim Exam				Done		
Proposal Defense					In Progress	
Dissertation					In Progress	

Figure 2. Timeline and current completion status of the seven case studies and major doctoral milestones.

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